$$E_1 = \frac{7}{4}\frac{ml^2}{T^2} \qquad E_2 = \frac{1}{4}\frac{ml^2}{T^2} \qquad E_3 = \frac{17}{36}\frac{ml^2}{T^2} \tag{9}$$

$$\frac{l}{v_n'} = \frac{n}{2}T \qquad n = 1,2,\cdots \tag{10}$$

$$v_n' = \frac{2l}{nT} \qquad n = 1,2,\cdots \tag{11}$$

$$\Delta t_{n1}' = \frac{n}{2}T \qquad \Delta t_{n2}' = \frac{n}{4}T \tag{12}$$

$$t_n' = \frac{T}{2} + \Delta t_{n1}' + \Delta t_{n2}' = \frac{3n+2}{4}T \tag{13}$$

$$t_n' \leqslant \frac{10}{4}T \tag{14}$$

$$n_{\max} = 2 \tag{15}$$

$$E_n' = \frac{1}{2}mv_2'^2 + E_0\sin\left[\left(\frac{2\pi}{T}\frac{3n+2}{4}\right)T + \frac{\pi}{4}\right]$$
$$n = 1,2 \tag{16}$$

$$E_1' = \frac{9}{4}\frac{ml^2}{T^2} \qquad E_2' = \frac{3}{4}\frac{ml^2}{T^2} \tag{17}$$

$$V_1' = \sqrt{\frac{9}{2}}\frac{l}{T} \tag{18}$$

$$V_2 = \sqrt{\frac{1}{2}}\frac{l}{T} \tag{19}$$

